

Regression Techniques

Lecture Notes

Indian Statistical Institute

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1 Lecture 01 : Introduction to Regression

1.1 Regression Setup

- Dependent variable $(y) \rightarrow Y$, we will work on data for example, AQI (UCI)
- Independent variables $(x_1, \dots, x_p) \rightarrow X_1, \dots, X_p$ independent variables e.g., NO, NO₂ etc., dependent variables example, temperature, humidity.

We assume

$$\mathbb{E}(Y) = g(x_1, \dots, x_p) \rightarrow \text{some function of } x_1, \dots, x_p \quad \text{Var}(Y) = \sigma^2$$

More precisely,

$$\mathbb{E}(Y \mid X_1 = x_1, \dots, X_p = x_p) = g(x_1, \dots, x_p)$$

Task of regression is to obtain $g(\cdot)$.

Possible distributions for Y : Normal, Exponential, Gamma.

1.2 Linear Regression

Linear regression:

$$g(x_1, \dots, x_p) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

Task: estimate $\beta_0, \dots, \beta_p$ based on data. These are parameters.

Definition 1 (Linear in parameters). *A linear regression model is a model which is linear in the parameters.*

Example: $\beta_0 + \beta_1 \sin x_1 + \beta_2 e^{x_2}$ is also a linear model (linear in β).

Thus

$$\mathbb{E}(Y) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p, \text{ is also a linear model}$$

Let $i = 1, \dots, n$ denote n time points. For the i -th observation:

$$\mathbb{E}(Y_i) = \beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi} \quad (\text{e.g., } Y_i \text{ is temp at time } i, x_{1i} \text{ is NO at time } i, \dots, x_{pi} \text{ is NO}_2 \text{ at time } i)$$

$$\text{Var}(Y_i) = \sigma^2, \quad i = 1, 2, \dots, n$$

Intuition: For each fixed $\tilde{x}$, $g(\tilde{x})$ gives mean of Y . Y_1, Y_2 distributions centered at $\mu_{Y_1} = E(Y_1)$, $\mu_{Y_2} = E(Y_2)$ with variance σ^2 . Distributions are independent.

1.3 Error Form and Matrix Form

We can write:

$$Y_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi} + \epsilon_i, \quad i = 1, \dots, n$$

$$\mathbb{E}(\epsilon_i) = 0, \quad \text{Var}(\epsilon_i) = \sigma^2$$

Indeed,

$$\mathbb{E}(Y_i) = \mathbb{E}(\beta_0 + \dots + \beta_p x_{pi} + \epsilon_i) = \beta_0 + \dots + \beta_p x_{pi} + \mathbb{E}(\epsilon_i) = \beta_0 + \dots + \beta_p x_{pi}$$

$$\text{Var}(Y_i) = \text{Var}(\epsilon_i)$$

Recall $\text{Var}(a + bX) = b^2 \text{Var}(X)$.

Let $x_1 = \text{NO}$, $x_p = \text{NO}_2$, etc.

Define random vector:

$$Y_n = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad X_n = \begin{bmatrix} 1 & x_{11} & \cdots & x_{p1} \\ 1 & x_{12} & \cdots & x_{p2} \\ \vdots & \vdots & & \vdots \\ 1 & x_{1n} & \cdots & x_{pn} \end{bmatrix}_{n \times (p+1)} \quad (\text{Design matrix})$$

$$\underset{\sim}{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} \quad (\text{regression coefficient}), \quad \underset{\sim}{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$$\mathbb{E}(\epsilon_i) = 0, \quad \text{Var}(\epsilon_i) = \sigma^2, \quad \epsilon_1, \dots, \epsilon_n \text{ are ind.}$$

The k th column of X_n is n observations on the k th dependent variable.

The m th row of X_n represents values of all the dependent variables at m th instance. Then model:

$$Y_n = X_n \underset{\sim}{\beta} + \underset{\sim}{\epsilon}$$

Properties:

$$\mathbb{E}(Y_n) = \begin{bmatrix} \mathbb{E}(Y_1) \\ \vdots \\ \mathbb{E}(Y_n) \end{bmatrix} = X_n \underset{\sim}{\beta} + \mathbb{E}(\underset{\sim}{\epsilon}) = X_n \underset{\sim}{\beta}$$

$$\text{Var}(Y_n) = \text{Var}(\underset{\sim}{\epsilon}) = \mathbb{E}(\underset{\sim}{\epsilon} \underset{\sim}{\epsilon}^T) = \sigma^2 I_n, \quad I_n = \text{Diag}(1, \dots, 1)_{n \times n}$$

1.4 Multivariate Notation Prerequisites

Let $\underset{\sim}{Z}_k = [Z_1, \dots, Z_k]^T$.

$$\mathbb{E}(\underset{\sim}{Z}_k) = \begin{bmatrix} \mathbb{E}(Z_1) \\ \vdots \\ \mathbb{E}(Z_k) \end{bmatrix}, \quad \text{Var}(\underset{\sim}{Z}_k) = \mathbb{E}[(\underset{\sim}{Z}_k - \mathbb{E}\underset{\sim}{Z}_k)(\underset{\sim}{Z}_k - \mathbb{E}\underset{\sim}{Z}_k)^T] = ((\text{Cov}(Z_i, Z_j)))_{k \times k}$$

Let $\tilde{X} = [X_1, \dots, X_p]^T$, $\tilde{Z} = [Z_1, \dots, Z_q]^T$.

$$\text{Cov}(\tilde{X}, \tilde{Z})^{p \times q} = ((\text{Cov}(X_i, Z_j)))$$

When $\mathbb{E}(\tilde{X}) = \mu_{\tilde{X}}$, $\text{Var}(\tilde{X}) = \sigma^2 I$.

Exercises:

1. $\text{Cov}(a^T \tilde{Z}, b^T \tilde{W}) =$
2. $\text{Cov}(\tilde{X}_1 + \tilde{X}_2, \tilde{Z}_1 + \tilde{Z}_2) =$
3. $\text{Var}(\tilde{X} + \tilde{Z}) =$
4. $\mathbb{E}(\tilde{X}^T \tilde{A} \tilde{X}) =$, where $\tilde{A}$ is $n \times n$.

2 Lecture 02: Mathematical Prerequisites and Continuation

We continue with model:

$$Y_i \stackrel{\text{iid}}{\sim} F \text{ with } \mathbb{E}(Y_i) = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi}, \quad \text{Var}(Y_i) = \sigma^2$$

$$Y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi} + \epsilon_i; \quad \epsilon_i \stackrel{\text{iid}}{\sim} G, \quad \mathbb{E}(\epsilon_i) = 0 \quad \forall i, \quad \text{Var}(\epsilon_i) = \sigma^2 \quad \forall i$$

2.1 Prerequisite 1: Gradient and Hessian

Let $g : \mathbb{R}^k \rightarrow \mathbb{R}$, e.g., $k = 2$, $g\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = x + y$.

In general $g(\underset{\sim}{x}) \in \mathbb{R}$ where $\underset{\sim}{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix}$.

Example: $g(\underset{\sim}{x}) = x_1 + \cdots + x_k$.

- Gradient:

$$\nabla g = \frac{\partial g}{\partial \underset{\sim}{x}} = \begin{bmatrix} \partial g / \partial x_1 \\ \vdots \\ \partial g / \partial x_k \end{bmatrix}^{k \times 1}$$

For $g(\underset{\sim}{x}) = x_1 + \cdots + x_k$,

$$\nabla g = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \underset{\sim}{1}$$

- Second derivative / Hessian:

$$\frac{\partial^2 g}{\partial \underset{\sim}{x} \partial \underset{\sim}{x}^T} = \frac{\partial}{\partial \underset{\sim}{x}} \begin{bmatrix} \frac{\partial g}{\partial x_1} & \cdots & \frac{\partial g}{\partial x_k} \end{bmatrix} = \begin{bmatrix} \frac{\partial^2 g}{\partial x_1^2} & \cdots & \frac{\partial^2 g}{\partial x_1 \partial x_k} \\ \frac{\partial^2 g}{\partial x_2 \partial x_1} & \cdots & \vdots \\ \vdots & \vdots & \vdots \\ \frac{\partial^2 g}{\partial x_k \partial x_1} & \cdots & \frac{\partial^2 g}{\partial x_k^2} \end{bmatrix}^{k \times k} = H(\underset{\sim}{x})$$

This is the **Hessian matrix** $H(\underset{\sim}{x}) = \left(\left(\frac{\partial^2 g}{\partial x_i \partial x_j} \right) \right)$.

- For vector-valued $g : \mathbb{R}^k \rightarrow \mathbb{R}^k$, $g(\underset{\sim}{x}) = \begin{bmatrix} g_1(\underset{\sim}{x}) \\ \vdots \\ g_k(\underset{\sim}{x}) \end{bmatrix}$:

$$\nabla g = \left[\left(\frac{\partial g_1}{\partial \underset{\sim}{x}} \right) \cdots \left(\frac{\partial g_k}{\partial \underset{\sim}{x}} \right) \right]^{k \times k} = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial g_k}{\partial x_1} \\ \vdots & \vdots & \vdots \\ \frac{\partial g_1}{\partial x_k} & \cdots & \frac{\partial g_k}{\partial x_k} \end{bmatrix}$$

Example: $g\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}$, then

$$\nabla g = \begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}$$

2.2 Use 1: Jacobian of Transformation

Let $\tilde{Z} = h(\tilde{X})$.

Recall the transformation technique of obtaining the distribution of a one-to-one function of X , when the distribution of X is known.

Univariate: $X \sim f_X(x)$ (pdf), $Z = h(X)$, then

$$f_Z(z) = f_X(x) \left| J_{x \rightarrow z} \right|_{x \rightarrow h^{-1}(z)}, \quad J_{x \rightarrow z} = \left| \frac{\partial x}{\partial z} \right|$$

Multivariate: $\tilde{X} \sim f_{\tilde{X}}$ (joint pdf), $\tilde{Z} = h(\tilde{X})$ where $h : \mathbb{R}^p \rightarrow \mathbb{R}^p$ is one-to-one function. The pdf of $\tilde{Z}$ at $\tilde{z}$ is:

$$f_{\tilde{Z}}(\tilde{z}) = f_{\tilde{X}}(\tilde{x}) \left| J_{\tilde{x} \rightarrow \tilde{z}} \right|_{\tilde{x} \rightarrow h^{-1}(\tilde{z})}, \quad J_{\tilde{x} \rightarrow \tilde{z}} = \left| \det \left(\frac{\partial \tilde{x}}{\partial \tilde{z}} \right) \right|$$

Example 1. Let $X_i \stackrel{iid}{\sim} N(0, \sigma^2), i = 1, 2$. Let $\tilde{Z} = \begin{pmatrix} X_1 + X_2 \\ X_1 - X_2 \end{pmatrix}$, $\tilde{Z} = A\tilde{X}$, $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.

$$f_{\tilde{X}}(\tilde{x}) = \frac{1}{2\pi\sigma^2} \exp \left\{ -\frac{1}{2\sigma^2} (x_1^2 + x_2^2) \right\}$$

We have $\tilde{X} = \begin{bmatrix} (z_1 + z_2)/2 \\ (z_1 - z_2)/2 \end{bmatrix} = B\tilde{Z}$,

$$J_{\tilde{x} \rightarrow \tilde{z}} = \left| \det \left(\frac{\partial \tilde{x}}{\partial \tilde{z}} \right) \right| = \left| \det \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \right| = \frac{1}{2}$$

Thus,

$$f_{\tilde{Z}}(\tilde{z}) = \frac{1}{2\pi\sigma^2} \exp \left\{ -\frac{1}{2\sigma^2} \left(\frac{(z_1 + z_2)^2}{4} + \frac{(z_1 - z_2)^2}{4} \right) \right\} \times \frac{1}{2}$$

$$z_1 \in \mathbb{R}, z_2 \in \mathbb{R}, \sigma^2 > 0$$

Identify the distribution of $\tilde{Z}$.

2.3 Use 2: Finding Maxima/Minima of $\mathbb{R}^k \rightarrow \mathbb{R}$ Function

Let $g : \mathbb{R}^k \rightarrow \mathbb{R}$, e.g., $g \left(\begin{pmatrix} x \\ y \end{pmatrix} \right) = x^3 + 3x^2y + y^3$.

Let $\tilde{z}_0$ be a maxima of g which is also an interior point of the domain of g .

To obtain $\tilde{z}_0$, one may follow steps below:

- Find ∇g and solve $\nabla g = 0$ (a set of k equations).
- Suppose $x_1^*, \dots, x_m^*$ are solutions to $\nabla g = 0$. Then these are called **Critical / Stationary points**. A critical point may be a maxima or minima or none of these.
- Find the Hessian at $x_1^*, \dots, x_m^*$, i.e., $H(x_1^*), \dots, H(x_m^*)$. If $H(x_i^*)$ is positive definite (negative definite) then x_i^* is a local minima (maxima).

Example 2. $g : \mathbb{R}^k \rightarrow \mathbb{R}$, $g(x) = \sum_{j=1}^k (x_j^4 - 4x_j^2)$.

$$i) \nabla g = \begin{bmatrix} 4x_1^3 - 8x_1 \\ \vdots \\ 4x_k^3 - 8x_k \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

Solutions: $x_j \in \{0, \sqrt{2}, -\sqrt{2}\}$. So total 3^k critical points:

$$x_{\sim 1}^* = 0, x_{\sim 2}^* = \sqrt{2}1, x_{\sim 3}^* = \begin{pmatrix} \sqrt{2} \\ \sqrt{2} \\ \vdots \\ -\sqrt{2} \end{pmatrix}, \dots, x_{\sim m}^* = -\sqrt{2}1, x_{\sim m+1}^* = \begin{pmatrix} 0 \\ \sqrt{2} \\ \vdots \\ \sqrt{2} \end{pmatrix}, \dots$$

$$ii) H(x) = \text{Diag}(12x_1^2 - 8, \dots, 12x_k^2 - 8)$$

$H(0) = -8I$: negative definite. So 0 is a local maxima.

For all combinations with $x_j = \pm\sqrt{2} \forall j$,

$$H = 16I : \text{positive definite}$$

So all these points are local minima.

Note: Points with mix of 0 and $\pm\sqrt{2}$ give indefinite Hessian.

Pre-requisite II : Multivariate Normal Distribution

Def: $X_{\sim}^{k \times 1}$ is said to follow a multivariate normal distr. with parameters $\underline{\mu} \in \mathbb{R}^k$ and $\underline{\Sigma}$ (p.s.d. matrix) if for any $\underline{a} \in \mathbb{R}^k$,

$$\underline{a}^T X_{\sim} \sim \mathcal{N}(\underline{a}^T \underline{\mu}, \underline{a}^T \underline{\Sigma} \underline{a}) \quad \checkmark$$

$$\left[\begin{array}{c} e.g. : \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \frac{\underline{e}_i^T X_{\sim} \sim \mathcal{N}(\mu_i, \sigma_{ii})}{X_i \sim \mathcal{N}(\mu_i, \sigma_{ii})} \quad \underline{\mu} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_k \end{pmatrix} \quad \underline{\Sigma} = ((\sigma_{ij})) \end{array} \right]$$

$$\begin{array}{c} X_1 \sim \mathcal{N} \\ X_2 \sim \mathcal{N} \\ \text{Cov}(X_1, X_2) = 0 \end{array} \not\Rightarrow \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N} \quad \Bigg| \quad \text{and Cov}(X_1, X_2) = 0$$

$$\left| f(x, y) = \frac{1}{2\pi} e^{-\left(\frac{x^2}{2} + \frac{y^2}{2}\right)} \left\{ 1 + xy e^{-\left(\frac{x^2}{2} + \frac{y^2}{2}\right)} \right\} \right|$$

Properties

1) Characteristic function

Univariate

$$\Phi_X(t) = E(e^{itX})$$

$$\Phi_{X_{\sim}}(\underline{t}) = E\left(e^{i\underline{t}^T X_{\sim}}\right)$$

Recall, $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$\Phi_X(t) = \exp\left\{i\mu t - \frac{1}{2}\sigma^2 t^2\right\}$$

What is $\Phi_{X_{\sim}}(\underline{t})$? $X_{\sim} \sim \mathcal{N}(\underline{\mu}, \underline{\Sigma})$

$$\begin{aligned} \Phi_{X_{\sim}}(\underline{t}) &= E\left(\exp\left\{i\underline{t}^T X_{\sim}\right\}\right) = E(\exp\{iW\}) \quad (W = \underline{t}^T X_{\sim} \sim \mathcal{N}(\underline{t}^T \underline{\mu}, \underline{t}^T \underline{\Sigma} \underline{t})) \\ &= \Phi_W(1) = \exp\left\{i\underline{t}^T \underline{\mu} - \frac{1}{2}\underline{t}^T \underline{\Sigma} \underline{t}\right\} \end{aligned}$$

$$\begin{aligned}
[2] \quad & \underset{\sim}{X}^{k \times 1} \sim \mathcal{N}(\underline{\mu}, \underline{\underline{\Sigma}}) \\
& \underset{\sim}{Y}^{m \times 1} = A \underset{\sim}{X}^{k \times 1} + \underline{b} \quad A \text{ is a full rank matrix} \\
& \Phi_{\underset{\sim}{Y}}(\underline{t}) \quad \left(\text{Use } \Phi_{\underset{\sim}{X}}(\underline{t}) = \exp \left\{ i \underline{\mu}^T \underline{t} - \frac{1}{2} \underline{t}^T \underline{\underline{\Sigma}} \underline{t} \right\} \right) \\
& \text{then find } \underset{\sim}{Y} \sim ?? .
\end{aligned}$$